

Deviations in Cropland Protection: The Multi-tasking Pressures by Local Bureaucrats in China

Research Proposal

Presented by Emily WANG

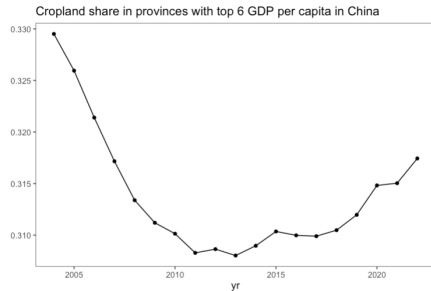
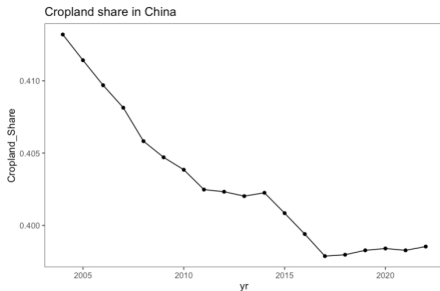
University of Tokyo

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Motivation

► "Red Line" Policy in China:

- China's total arable land shall be no less than 1.8 billion *mu* (120 million hectares)
- 2005: The State Council released assessment measures for provincial governments¹
- 2018: A revised version has been introduced to assess local land protection more stringently²



¹Document of 2005 policy

²Document of 2018 policy

Motivation

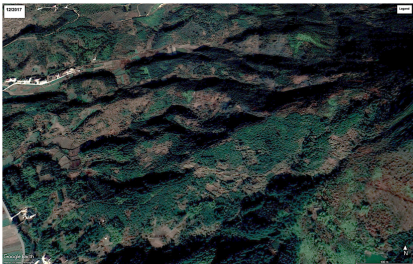


Figure: An example of local policy implementation: forest → terrace

► **Anecdotes on deviation in local enforcement:**

- Green for grain, Rice on mountain
- Allocative inefficiency

Research Overview

- ▶ **RQ:** (Under a target-based performance evaluation system in which both land protection (2018 -) and economic growth (*always*) are counted, will the multitasking local officials deviate their local-level regulatory enforcement to favor economic objectives and thus result in land allocation inefficiency?)
- ▶ **Hypothesis:**
 - ▶ **H1:** The land resource-constrained prefectures react more to the land protection assessment policy and witness a larger increase in cropland share after 2018.
 - ▶ **Outcome of interest:** cropland share (grid/county)
 - ▶ **Treatment intensity:** although the land policy is nationwide, the intensity of the policy hinges on the (1) proportion of available land and (2) urban density level in each region. ▶ Policy
 - ▶ ▶ Preliminary results

Research Overview

Hypothesis:

- ▶ **H1:** The land resource-constrained prefectures react more to the land protection assessment policy and witness a larger increase in cropland share after 2018.
- ▶ **H2:** After the implementation of the 2017 policy, in prefectures where land resources are scarce and the provincial leader has stronger incentives to fulfill the regulation, newly reclaimed land will have lower land productivity and agricultural output.
 - ▶ **Proxy of land resource scarcity:** available land share in the prefecture (1 - urban district - high altitude)
 - ▶ **Proxy of land productivity:** no direct measurement; soil properties, Topographic data (altitude), mudslide/landslide data
 - ▶ **Proxy of land productivity:** no cropland-level data, only county-level data (self-reported agricultural output at the county level)

Contribution to literature

► Land regulation:

- housing prices (Hilber and Vermeulen, 2016), economic growth (Besley and Burgess, 2000), farmer-level productivity (Beg, 2022; Adamopoulos et al., 2022 ([resource misallocation across farmer; my study: misallocation across land](#)))
- Chinese context: Henderson et al. (2022), Cai et al. (2017), Adamopoulos (2022), Cheng and Chung (2018)

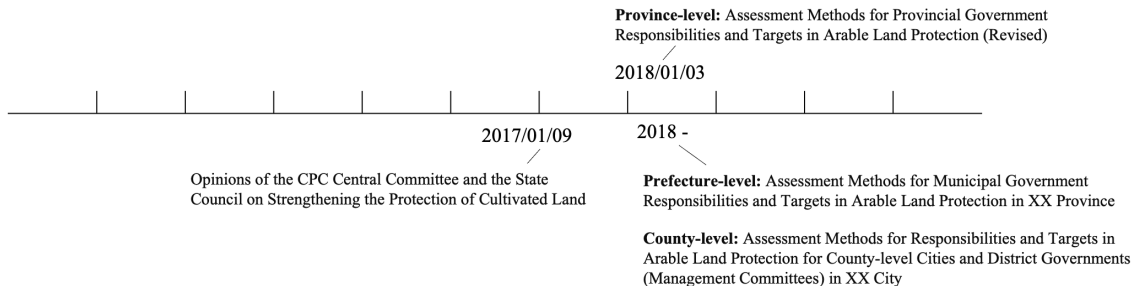
► Political economy in a centralized regime:

- Theory: principal-agent problem (Ross, (1973)), overcentralization (Kornai, (1959)), multi-tasking (Holmstrom and Milgrom (1991), Hart, Shleifer, and Vishny (1997))
- Empirical studies: environmental regulation (Fan et al., (2023), Chen et al., (2018), He et al., (2020))

► Contribution:

- multitasking bureaucrats balance economic objectives and ([previous literature](#)) - [environmental regulation/land regulation in urban area](#), ([my study](#)) - [land regulation](#)
- cropland protection regulation-induced cropland misallocation

Timeline

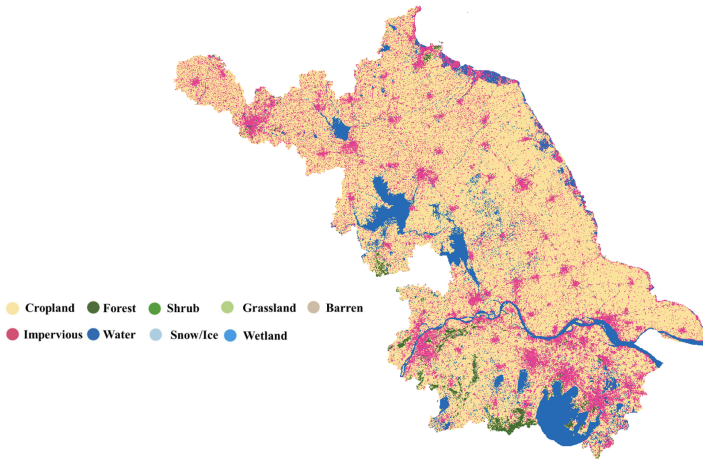


Data

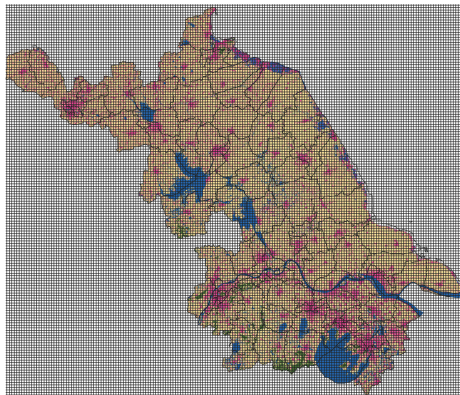
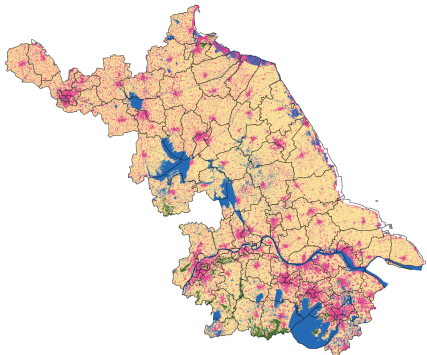
- ▶ Cropland data:
 - ▶ Raster data: The 30 m annual land cover dataset and its dynamics in China from 1990 to 2022 ([now using 2011 - 2022](#))
 - ▶ County-level data: Statistical yearbook of China
- ▶ 2010 census data: [info on urban density, population, GDP...](#)
- ▶ Soil property data: cross-section
- ▶ Topographical data: cross-section
- ▶ Bureaucrats data: prefecture-, and province-level officials (gender, age, education, ethnicity, hometown, and tenure) from 1992 - 2022
- ▶ Old administrative division (Qing dynasty), location of historical villages

Preliminary analysis on cropland share

- ▶ A small-scale analysis in one province - Jiangsu province: (1) top-ranked GDP per capita, (2) covered by plain

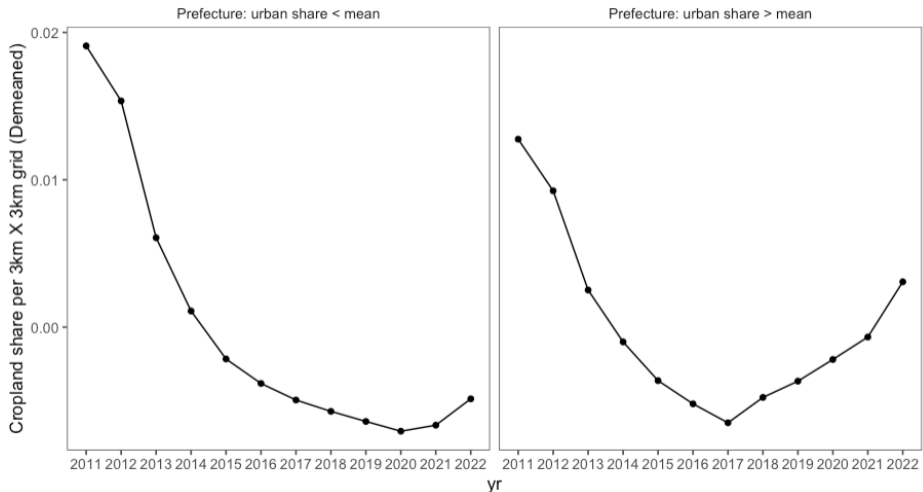


Preliminary analysis on cropland share



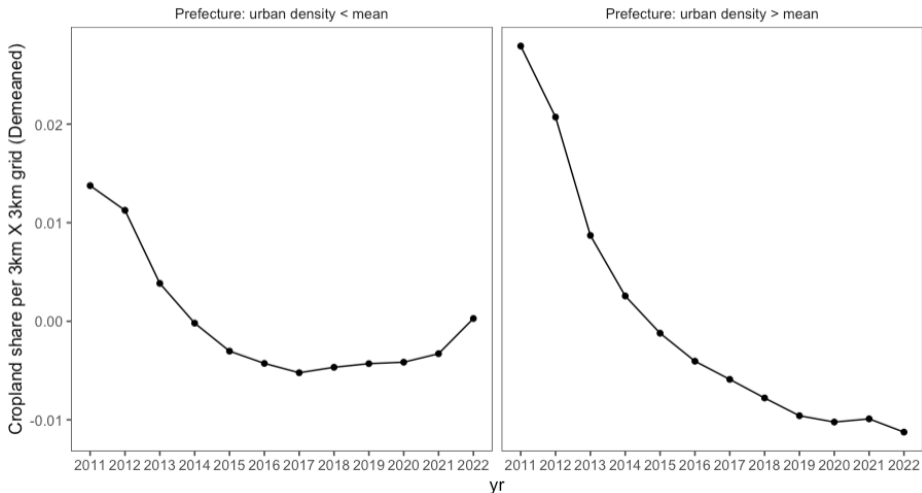
Descriptive statistics: high urban share v.s. low urban share

Cropland change in Jiangsu province



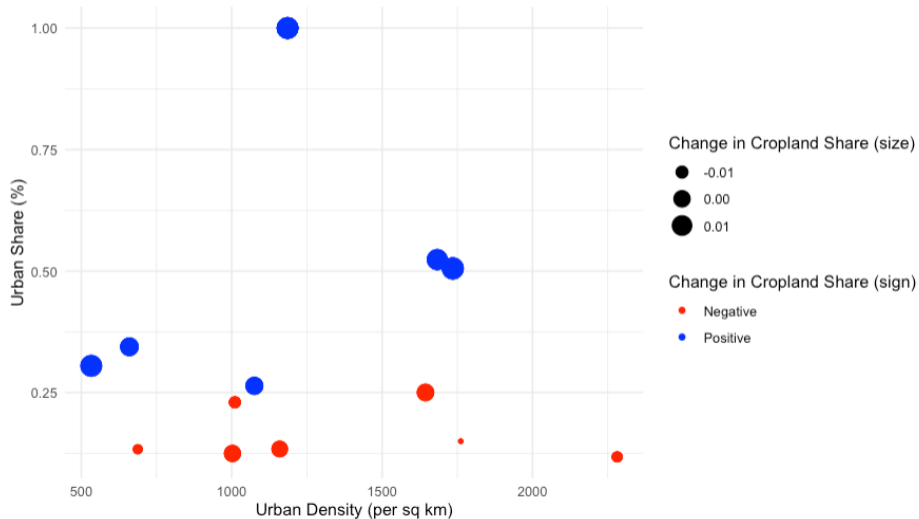
Descriptive statistics: high urban density v.s. low urban density

Cropland change in Jiangsu province



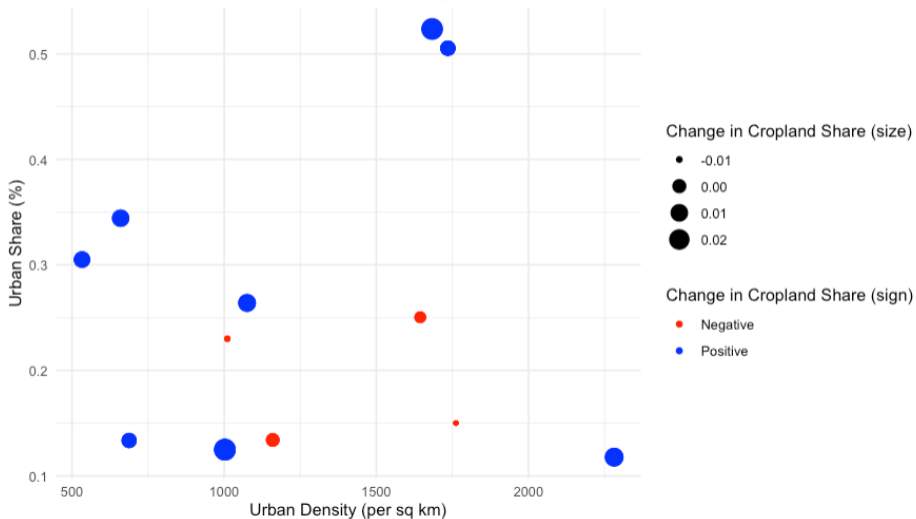
Descriptive statistics: cropland in Jiangsu province

Cropland Share in urban districts: Comparing 2017 and 2022



Descriptive statistics: cropland in Jiangsu province

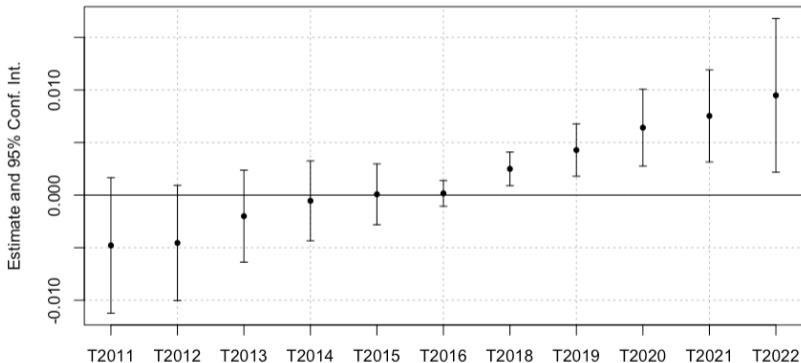
Cropland Share in rural districts: Comparing 2017 and 2022



Naive regression: event study plot

$$y_{ict} = \sum_{t \in \{2011, \dots, 2016, 2018, \dots, 2022\}} \gamma_t \cdot T_{ict} + \alpha_c + \delta_t + \epsilon_{ict}$$

Effect on cropland share per grid ($T = 1\{\text{urban share} > \text{mean}\}$)

[Return](#)

Cropland policy intensity

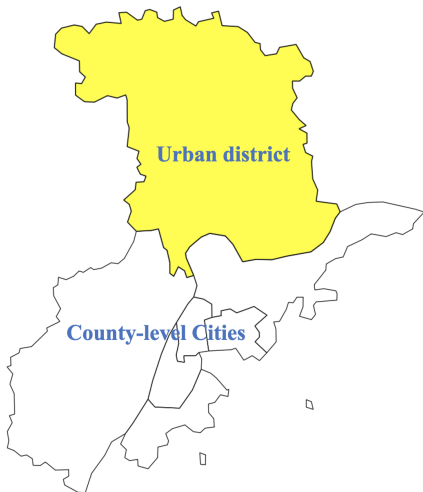
► Cropland protection policy is binding more on prefectures with

1. Higher in urban density: urbanization, economic growth
2. Higher **urban share** in prefecture

Cropland balance policy *"If they (local governments) are unable to independently compensate with an equivalent amount and quality of arable land, they must pay the arable land reclamation fee in full as required. Local governments at all levels are responsible for organizing and implementing land consolidation ... The primary method is (1) self-balancing within counties, (2) supplemented by adjustments within provinces, and (2) further supplemented by appropriate national coordination, to fulfill the task of compensating for arable land."*

Example on high urban share v.s. low urban share

Prefecture 1



Prefecture 2

